and more than fifty years of continued experiments have failed to find any cracks in it wide enough to stick a new particle through.

But CERN keeps trying. And not just because smashing things is, admittedly, quite a lot of fun.

The name of the game in particle colliders is energy. Throwing particles at each other faster means the eventual collision is at a higher energy, and the higher energy your collisions, the larger the swath of possible new physics you can reach. You can think of collision energy as legal tender, to be exchanged, via $E = mc^2$, for particle mass. If the total energy in a collision is higher than the equivalent mass of the particle you're trying to create, then as long as your theory allows *any* kind of interaction between that particle and the ones you've smashed together, you have a chance of creating that particle. Extensions to the Standard Model tend to involve particles that are significantly heavier than those we've detected so far, which means that we need to reach higher and higher energies to find them. But even when you've reached the right energy threshold, it takes more than one creation of a particle to get a meaningful, statistically significant signal. The Large Hadron Collider had to run for years, smashing countless trillions of protons,[fn5] before it had collected enough data to say with acceptable certainty that a Higgs boson had been found.

It's this constant push into the energy frontier that leads to CERN's unfortunate reputation as an existential menace. The thinking goes that if humanity has never before seen so much energy concentrated into one place, *who knows what could happen?* Some of the concerns include the unsettling scenarios we've discussed in previous chapters, like the creation of little black holes, or the triggering of catastrophic vacuum decay. Fortunately, in every disaster scenario presented so far, we can easily set aside the worries based on the fact that the LHC is hardly even a blip compared to the particle-obliterating violence going on all around us in the universe. But in the minds of certain especially fretful nonphysicists, not every worry is as well defined, or as easily assuaged, even though the LHC has operated totally harmlessly for over a decade. By the time I visited CERN in February 2019, internet jokes about the LHC tearing open a portal into another dimension, or shifting the universe into "the bad timeline," seemed about as prevalent as ever.